Safety of MRAC + min-norm CBF augmentation with an unknown A matrix: is the tube necessary?

Notes for `Robust_Adaptive.m` and `Robust_Adaptive_TK.m` (lateral-directional aircraft example, Menner & Lavretsky). October 2026.

Short answers.

1. **Your reasoning is correct.** While the gains are adapting, the aircraft does not behave like any model the controller can compute. A barrier condition evaluated without a margin therefore cannot guarantee the original bounds. In my earlier message I said the tube is not needed for `Robust_Adaptive_TK.m`. That was too strong: that design gives a *bounded and vanishing* violation (Theorem 1), not a guarantee. The β peak of 0.5005° it produced is exactly such a violation.
2. **Your tube gives a hard guarantee** (Theorem 2), with one refinement to the picture. The CBF acts on the *reference model*, which is known exactly, so the reference model never violates the shrunken bounds. The *plant* may cross the shrunken bounds by up to $\delta_i(t)$, the full tube width rather than just "minimally", but it can never cross the original bounds.
3. **The guarantee needs three things:** a known bound $\theta_{\max}$ on the uncertainty; a non-empty shrunken box (δ_i smaller than the half-width of the bounds); and $x(0) = x_{\mathrm{ref}}(0)$. Clamping the tube, as in `min(delta, 0.9*Y_max)`, voids the guarantee.
4. **Why your tube looked unusable.** With $\gamma = 10^4$ and the open-loop reference model, the tube at $t = 0$ is 3.99° for the rudder (limit 2°) and 0.78° for β (limit 0.5°). The shrunken box was empty. The tube width scales like $\theta_{\max}/\sqrt{\gamma}$. With the observer-like reference model, a larger γ is affordable: at $\gamma = 10^5$ the tube is $[0.29^\circ,\ 0.65^\circ,\ 0.27\,\mathrm{dps},\ 0.26^\circ]$, which is feasible (Section 6).
5. **Safety can be guaranteed without shrinking the bounds** by putting the margin on the barrier's *derivative* instead (Proposition 3, a robust CBF). Some margin, however, is unavoidable (Remark 4). For this aircraft the derivative-margin version is infeasible on the rudder row, so your tube is the right tool here.

1. Setup

The extended state is $x = [e_{yI}^\top\ x_p^\top]^\top \in \mathbb{R}^n$ ($n = 5$, $m = 2$). The true plant has an unknown matrix $\Theta \in \mathbb{R}^{m \times n}$ that enters through the known input matrix B (matched uncertainty). In the code this is `A_true = A + B*Theta`:

$$\dot{x} = (A + B\Theta)\,x + B\,u + B_{\mathrm{ref}}\,y_{\mathrm{cmd}} + B_v\,v, \qquad B_{\mathrm{ref}} = -B_v.$$

A, B, B_v are known. The baseline LQR gain gives the Hurwitz matrix $A_{\mathrm{ref}} = A - BK_{\mathrm{lqr}}$, and the *ideal* gain $K^\star = K_{\mathrm{lqr}} + \Theta$ satisfies $A + B\Theta - BK^\star = A_{\mathrm{ref}}$. The controller and the reference model are

$$u = -\hat{K}x + w, \qquad \dot{x}_{\mathrm{ref}} = A_{\mathrm{ref}}x_{\mathrm{ref}} + B_\pi(\pi_0 + \pi) + L_v\,(x - x_{\mathrm{ref}}),$$

with $B_\pi = [B_v \;\; B]$, $\pi = [v^\top \; w^\top]^\top$ the CBF signals, and $\pi_0 = [-y_{\text{cmd}}^\top \; 0]^\top$, so that $B_\pi \pi_0 = B_{\text{ref}} y_{\text{cmd}}$. The original file uses $L_v = 0$ (open-loop reference model). The TK file uses $L_v = P_v R_v^{-1}$ (observer-like reference model). Define $e = x - x_{\text{ref}}$, $\Delta K = \hat{K} - K^\star$ and $A_e = A_{\text{ref}} - L_v$ (Hurwitz). Let $P = P^\top \succ 0$ satisfy $A_e^\top P + P A_e = -\tilde{Q}$ with $\tilde{Q} \succ 0$. The original file has $P =$ `lyap(A_ref',I)` and $\tilde{Q} = I$; the TK file has $P = P_v^{-1}$. The adaptive law, with $\Gamma = \text{blkdiag}(\Gamma_I, \Gamma_P) \succ 0$, is

$$\dot{\hat{K}} = B^\top P \, e \, x^\top \Gamma.$$

Constrained outputs. $y_{\text{lim}} = C_{\text{lim}}(t)\, x$ with $C_{\text{lim}}(t) = \begin{bmatrix} -\hat{K}(t) \\ 0 \;\; C_{p\,\text{lim}} \end{bmatrix}$. The rows $c_i(t)$, $i = 1, \dots, 2m$, are: $u_{\text{bl}} = -\hat{K} x$ (aileron, rudder), then p_s and β. Bounds hold componentwise, $y^{\min} \le y_{\text{lim}} \le y^{\max}$ with $y_i^{\min} < y_i^{\max}$. All rows are treated as relative degree one with slopes $\alpha = \text{diag}(\alpha_i) \succ 0$, and

$$H_x = C_{\text{lim}} A_{\text{ref}} + \alpha C_{\text{lim}}, \qquad H_u = C_{\text{lim}} B_\pi = \begin{bmatrix} -\hat{K}_I & -\hat{K}_I D_{p\,\text{reg}} - \hat{K}_P B_p \\ 0 & C_{p\,\text{lim}} B_p \end{bmatrix}.$$

Assumptions. (A1) Matching: the uncertainty is $B\Theta$ with B known. (A2) Prior bound: $\|K^\star - \hat{K}(0)\|_F = \|\Theta\|_F \le \theta_{\max}$ (code: $\hat{K}(0) = K_{\text{lqr}}$, $\theta_{\max} = 1.5$; true value 1.3625). (A3) $x(0) = x_{\text{ref}}(0)$, so $e(0) = 0$. (A4) $H_u(t)$ is invertible ($\hat{K}_I(t)$ and $C_{p\,\text{lim}} B_p$ invertible). (A5) Solutions exist on $[0, \infty)$. The closed loop is locally Lipschitz because the CBF law is continuous and piecewise linear.

2. Three lemmas

> **Lemma 1 (closed-form CBF satisfies every modified constraint).** Let $a, b \in \mathbb{R}^{2m}$ with $a_i + b_i < 0$ for every i, and $\pi = H_u^{-1}\big(\max(0, b) - \max(0, a)\big)$. Then $a + H_u \pi \le 0$ and $b - H_u \pi \le 0$ componentwise.

Proof. $(H_u \pi)_i = \max(0, b_i) - \max(0, a_i)$. If $a_i > 0$, then $b_i < -a_i < 0$, so $(H_u \pi)_i = -a_i$, $a_i + (H_u \pi)_i = 0$ and $b_i - (H_u \pi)_i = a_i + b_i < 0$. The case $b_i > 0$ is symmetric. If $a_i, b_i \le 0$, then $(H_u \pi)_i = 0$. ■

In the code, `H_max_delta` plays the role of a and `H_min_delta` of b. Without tightening, $a_i + b_i = -\alpha_i(y_i^{\max} - y_i^{\min}) < 0$ always holds. With tightening, this becomes the feasibility condition of Theorem 2.

> **Lemma 2 (certainty-equivalence identity).** For any $\hat{K}$, the estimated plant $\hat{A} = A + B(\hat{K} - K_{\text{lqr}})$ satisfies $\hat{A} - B\hat{K} = A_{\text{ref}}$. The true closed loop satisfies $A + B\Theta - B\hat{K} = A_{\text{ref}} - B\,\Delta K$.

Proof. Direct substitution, using $\Theta = K^\star - K_{\text{lqr}}$. ■

This is why H_x is built from A_{ref} even though A is unknown. The only unknown left in the plant's barrier derivative is $-c_i B\,\Delta K\, x$.

Lemma 3 (MRAC Lyapunov function and the tube bound). Let $V = e^\top P e + \operatorname{tr}(\Delta K\, \Gamma^{-1} \Delta K^\top)$ and define the computable, non-increasing signal

$$\bar{V}(t) = \bar{V}_0 - \int_0^t e^\top \tilde{Q}\, e \, d\tau, \qquad \bar{V}_0 = \theta_{\max}^2\, \lambda_{\max}(\Gamma^{-1}) \quad (= \theta_{\max}^2/\gamma \text{ for } \Gamma = \gamma I).$$

Then, under (A1)–(A3), $0 \le V(t) \le \bar{V}(t)$ for all $t \ge 0$, and for every row vector c:

$$\text{(a)}\ \ |c\, e(t)| \le \sqrt{c\, P^{-1} c^\top}\, \sqrt{\bar{V}(t)}, \qquad \text{(b)}\ \ \|\Delta K(t)\|_F^2 \le \lambda_{\max}(\Gamma)\, \bar{V}(t).$$

Proof. By Lemma 2, $\dot{x} = (A_{\text{ref}} - B\Delta K)x + B_\pi(\pi_0 + \pi)$. Subtracting the reference model gives

$$\dot{e} = A_e\, e - B\, \Delta K\, x.$$

The CBF signals π cancel because they drive the plant and the reference model identically. The MRAC proof is therefore untouched by the CBF. Since $K^\star$ is constant, $\dot{\Delta K} = \dot{\hat{K}}$, and

$$\dot{V} = -e^\top \tilde{Q} e - 2 e^\top P B \Delta K x + 2\operatorname{tr}\big(\Delta K \Gamma^{-1} \Gamma x e^\top P B\big) = -e^\top \tilde{Q} e,$$

because $\operatorname{tr}(\Delta K x e^\top P B) = e^\top P B \Delta K x$. Hence $V(t) = V(0) - \int_0^t e^\top \tilde{Q} e$. With $e(0) = 0$, $V(0) = \operatorname{tr}(\Delta K(0) \Gamma^{-1} \Delta K(0)^\top) \le \lambda_{\max}(\Gamma^{-1}) \theta_{\max}^2 = \bar{V}_0$, so $V \le \bar{V}$. For (a), Cauchy–Schwarz in the P-norm gives $|ce| = |(cP^{-1/2})(P^{1/2} e)| \le \sqrt{cP^{-1}c^\top}\sqrt{e^\top P e} \le \sqrt{cP^{-1}c^\top}\sqrt{V}$. For (b), $\operatorname{tr}(\Delta K \Gamma^{-1} \Delta K^\top) \ge \|\Delta K\|_F^2 / \lambda_{\max}(\Gamma)$. ■

Your original code implements exactly this: `V_bar0 = theta_max^2/gamma`, `int_eQe`, and `delta = sqrt(diag(C_lim*P_inv*C_lim')*V_bar)`.

3. Without a margin: CBF on the plant state (what `Robust_Adaptive_TK.m` does)

Theorem 1 (bounded, vanishing violation; no guarantee). Suppose the CBF is evaluated on the plant state with the certainty-equivalence model:

$$a = H_x x + H_u \pi_0 + \dot{C}_{\text{lim}} x - \alpha y^{\max}, \qquad b = -H_x x - H_u \pi_0 - \dot{C}_{\text{lim}} x + \alpha y^{\min},$$

and π is given by Lemma 1. Let $d_i(t) = -c_i(t)\, B\, \Delta K(t)\, x(t)$, the model error that no measurement reveals. If $y^{\min} \le C_{\text{lim}}(0) x(0) \le y^{\max}$, then for all $t \ge 0$

$$c_i(t)x(t) \ \le\ y_i^{\max} + \int_0^t e^{-\alpha_i (t-\tau)} d_i(\tau)\, d\tau \ \le\ y_i^{\max} + \frac{1}{\alpha_i} \sup_{\tau \le t} \max\big(d_i(\tau), 0\big),$$

and the min side is symmetric. The violation is bounded by the model error divided by the slope (input-to-state safety). It is zero only if $c_i B\, \Delta K\, x \equiv 0$.

Proof. Let $h_i = c_i x - y_i^{\max}$. By Lemma 2, $\dot{h}_i = \dot{c}_i x + c_i A_{\text{ref}} x + c_i B_\pi(\pi_0 + \pi) - c_i B \Delta K x$, so $\dot{h}_i + \alpha_i h_i = a_i + (H_u \pi)_i + d_i \le d_i$ by Lemma 1. Since $h_i(0) \le 0$, the comparison lemma gives $h_i(t) \le \int_0^t e^{-\alpha_i(t-\tau)} d_i\, d\tau$. ■

Remarks. (i) If x is bounded, then $e \in L_2 \cap L_\infty$ and $\dot{e}, \ddot{e} \in L_\infty$. Barbalat's lemma then gives $e \to 0$ and $\dot{e} \to 0$, hence $B\Delta K x = A_e e - \dot{e} \to 0$ and $d_i \to 0$: the violation vanishes asymptotically. (ii) For the β row, $c_\beta B = [0 \;\; 0.0153]$, so d_β is tiny. This explains why TK held β to 0.5005° rather than exactly 0.5°. (iii) Theorem 1 is the honest statement for TK: good performance and small violations, but no guarantee.

4. With a tube: CBF on the reference model (your idea, made rigorous)

Define the tube and the shrunken bounds

$$\delta_i(t) = \sqrt{c_i(t) P^{-1} c_i(t)^\top \, \bar{V}(t)}, \qquad y_i^{\min} + \delta_i(t) \;\le\; c_i(t)\, x_{\text{ref}} \;\le\; y_i^{\max} - \delta_i(t).$$

The CBF is evaluated on x_{ref}. Every term of the reference model is known, including $L_v e$, which is measured. With

$$\ell_i = \big(H_x x_{\text{ref}} + H_u \pi_0 + \dot{C}_{\lim} x_{\text{ref}} + C_{\lim} L_v\, e\big)_i,$$
$$a_i = \ell_i + \dot{\delta}_i - \alpha_i\big(y_i^{\max} - \delta_i\big), \qquad b_i = -\ell_i + \dot{\delta}_i + \alpha_i\big(y_i^{\min} + \delta_i\big),$$

π from Lemma 1, and $u = -\hat{K}x + w$. Here $\dot{\delta}_i = \big(2\,\dot{c}_i P^{-1} c_i^\top \bar{V} - c_i P^{-1} c_i^\top \, e^\top \tilde{Q} e\big)/(2\delta_i)$ is computable, with $\dot{c}_i = -\dot{\hat{K}}_i$ for the input rows and $\dot{c}_i = 0$ for p_s and β.

Theorem 2 (guaranteed safety with unknown A). Assume (A1)–(A5) and the feasibility condition

$$\text{(F)} \qquad 2\,\delta_i(t) + \frac{2\,\dot{\delta}_i(t)}{\alpha_i} \;<\; y_i^{\max} - y_i^{\min} \quad \text{for all } t \ge 0 \text{ and all } i,$$

together with $x_{\text{ref}}(0)$ inside the shrunken box at $t = 0$. Then for all $t \ge 0$ and every i:

1. the reference model stays in the shrunken box: $y_i^{\min} + \delta_i(t) \le c_i(t) x_{\text{ref}}(t) \le y_i^{\max} - \delta_i(t)$;
2. the **plant stays in the original box**: $y_i^{\min} \le c_i(t) x(t) \le y_i^{\max}$. That is, $u_{\text{bl}} = -\hat{K}x \in [u^{\min}, u^{\max}]$, and p_s and β stay within their limits.

Proof. (1) Let $\tilde{h}_i = c_i x_{\text{ref}} - y_i^{\max} + \delta_i$. Differentiating along the reference model, which is known exactly,

$$\dot{\tilde{h}}_i + \alpha_i \tilde{h}_i = \dot{c}_i x_{\text{ref}} + c_i\big(A_{\text{ref}} x_{\text{ref}} + B_\pi(\pi_0 + \pi) + L_v e\big) + \dot{\delta}_i + \alpha_i \tilde{h}_i = a_i + (H_u \pi)_i.$$

Next, $a_i + b_i = 2\dot{\delta}_i - \alpha_i\big(y_i^{\max} - y_i^{\min} - 2\delta_i\big) < 0$ by (F), so Lemma 1 applies and $\dot{\tilde{h}}_i + \alpha_i \tilde{h}_i \le 0$. The comparison lemma gives $\tilde{h}_i(t) \le e^{-\alpha_i t}\tilde{h}_i(0) \le 0$. The min side is identical. No unknown quantity appears anywhere in this step, which is why it holds exactly.
(2) $c_i x = c_i x_{\text{ref}} + c_i e \le \big(y_i^{\max} - \delta_i\big) + \delta_i = y_i^{\max}$, by (1) and Lemma 3(a). The min side is identical. ∎

Remarks on Theorem 2.

- **Your intuition, made precise.** The shrunken box is enforced *exactly* on the reference model. The plant is only guaranteed to stay within δ_i of the reference model, so it may cross the shrunken bound by up

to δ_i, but never the original one.

- **Condition (F).** For p_s and β, $\dot{c}_i = 0$, so $\dot{\delta}_i \le 0$ and (F) reduces to $\delta_i(0) < (y_i^{\max} - y_i^{\min})/2$. For the u_{bl} rows, $\dot{\delta}_i$ can be positive while $\hat{K}$ moves, so (F) should be monitored online. Alternatively, replace $c_i P^{-1} c_i^\top$ by a constant upper bound; Lemma 3(b) bounds $\|\hat{K}(t) - \hat{K}(0)\|_F \le (1 + \sqrt{\kappa(\Gamma)})\theta_{\max}$. Then δ_i is non-increasing and (F) only needs to hold at $t = 0$, at the price of more conservatism.
- **The original code omitted $\dot{\delta}_i$.** For the output rows this is harmless because $\dot{\delta}_i \le 0$, which makes the condition conservative. For the input rows it introduces a small, unguarded term. It also omitted $C_{\text{lim}} L_v e$, which is zero there because $L_v = 0$.
- **What is guaranteed for u.** As in both papers, the bound is imposed on $u_{\text{bl}} = -\hat{K}x$. That the total command $u_{\text{bl}} + w$ stays in bounds relies on Theorem 3 of the ACC paper (w opposing u_{bl}), which is not re-proved here.
- **Price.** Conservatism: the reference model is held to $y^{\max} - \delta$. The tube shrinks only as $\sqrt{\bar{V}(t)}$, and $\bar{V}$ decreases only by the tracking-error energy. Since the gains need not converge (no persistent excitation), δ does not go to zero. Increasing γ is the main way to shrink it.

5. Without shrinking the bounds: a margin on the derivative

> **Proposition 3 (robust CBF on the plant).** Evaluate the CBF on x as in Theorem 1, but add to both a_i and b_i the computable margin
>
> $$\rho_i(t) = \|c_i(t)B\|_2 \sqrt{\lambda_{\max}(\Gamma)\,\bar{V}(t)}\;\|x(t)\|_2.$$
>
> If $2\rho_i(t) < \alpha_i\big(y_i^{\max} - y_i^{\min}\big)$ for all t, then $y_i^{\min} \le c_i(t)x(t) \le y_i^{\max}$ for all $t \ge 0$. The original bounds are not shrunk.

Proof. By Lemma 3(b), $|d_i| \le \|c_i B\|\,\|\Delta K\|_F\,\|x\| \le \rho_i$. The feasibility assumption makes Lemma 1 applicable to $(a + \rho,\ b + \rho)$, so $a_i + \rho_i + (H_u\pi)_i \le 0$. Hence $\dot{h}_i + \alpha_i h_i = a_i + (H_u\pi)_i + d_i \le d_i - \rho_i \le 0$. The min side is identical. ∎

For $\Gamma = \gamma I$, $\sqrt{\lambda_{\max}(\Gamma)\bar{V}_0} = \theta_{\max}$ regardless of γ. Unlike the tube, this margin does *not* shrink as adaptation gets faster. Take a representative $\|x\| = 0.3$ during the turn. With $\|c_i B\| = [2.82,\ 4.67,\ 8.41,\ 0.0153]$, the margins are $\rho \approx [1.27,\ 2.10,\ 3.78,\ 0.0069]$, against feasibility limits $\alpha_i y_i^{\max} = [4.19,\ 0.279,\ 12.6,\ 0.035]$. **The rudder row is infeasible.** In addition, the β margin is multiplied by about 65 in H_u^{-1}, which demands roughly 26° of rudder. For this aircraft, the tube is the better place to put the margin.

Remark 4 (some margin is unavoidable). Nagumo's condition is also *necessary*: a set $\{h \le 0\}$ is forward invariant only if $\dot{h} \le 0$ whenever $h = 0$. At a boundary point with $x \ne 0$, the true derivative equals the computable part plus $d_i = c_i B(K^\star - \hat{K})x$. This term is affine in the unknown $K^\star$. Unless $c_i B = 0$, some admissible $K^\star$ (within the $\theta_{\max}$ ball) makes $d_i > 0$. A controller that drives the computable part to exactly zero at the boundary, as the margin-free CBF does, can therefore be pushed out

by an admissible plant. To be safe for *every* admissible Θ, the controller needs a margin: either in the state (the tube) or in the derivative (Proposition 3). Both margins scale with $\theta_{\max}$. With no prior bound on Θ, no finite margin suffices, so no guarantee is possible. (This is an informal argument, not a theorem.)

6. Numbers for this aircraft

Table 1 lists the tube width at $t = 0$, $\delta_i(0) = \sqrt{c_i P^{-1} c_i^\top}\, \theta_{\max} / \sqrt{\gamma}$, with $\theta_{\max} = 1.5$ and c_i evaluated at $\hat{K}(0) = K_{\text{lqr}}$. A design is feasible only if $\delta_i(0)$ is below the limit, since the bounds are symmetric.

Design	**aileron (lim 3°)**	**rudder (lim 2°)**	p_s **(lim 18 dps)**	β **(lim 0.5°)**
Original: $L_v = 0$, $P = \text{lyap}$, $\gamma = 10^4$	1.105	**3.987**	2.614	**0.783**
Observer-like RM, $\gamma = 10^3$ (TK)	2.850	**6.540**	2.731	**2.601**
Observer-like RM, $\gamma = 10^4$	0.901	**2.068**	0.864	**0.823**
Observer-like RM, $\gamma = 10^5$	0.285	0.654	0.273	0.260
Observer-like RM, $\gamma = 3 \cdot 10^5$	0.165	0.378	0.158	0.150
Observer-like RM, $\gamma = 10^6$	0.090	0.207	0.086	0.082

Table 1. Tube width $\delta_i(0)$ in deg (dps for p_s). Red entries leave an empty shrunken box: the guarantee is impossible.

The original design was infeasible from the start, which is why the clamp to `0.9*Y_max` was needed. A clamp gives a tube that no longer bounds $|c_i e|$, so Theorem 2 no longer applies.

Numerical check of Theorem 2

Setup: the tube design of Section 4 (CBF on x_{ref}; $\delta_i(t)$ from $\bar{V}(t)$; $\dot{\delta}_i$ and $C_{\lim} L_v e$ included; observer-like RM with $\nu = 0.1$; $\alpha = \text{diag}(80, 8, 40, 4)$). Uncertain plant $\Theta = 0.5[0\ K_P]$, the same command as before, RK4 with $\Delta t = 10^{-4}$. Each line checks one claim of the theorem.

Check (deg; dps for p_s)	**Tube, $\gamma = 10^5$**	**Tube, $\gamma = 3 \cdot 10^5$**	**No tube, $\gamma = 10^3$**	**No tube, $\gamma = 10^5$**
Tube width $\delta(0)$ [ail, rud, p_s, β]	[0.29, 0.65, 0.27, 0.26]	[0.17, 0.38, 0.16, 0.15]	0	0
Claim 1: $\max_t \left(\lvert c_i x_{\text{ref}}\rvert - (y_i^{\max} - \delta_i)\right) \le 0$	holds (β row on boundary)	holds	holds	holds
Lemma 3(a): $\max_t \left(\lvert c_i e\rvert - \delta_i\right) \le 0$	[−0.26, −0.62, −0.26, −0.25]	[−0.15, −0.36, −0.15, −0.15]	n/a (actual $\lvert c_i e\rvert$ up to [0.05, 0.23, 0.07, 0.027])	n/a (actual $\lvert c_i e\rvert$ up to [0.02, 0.06, 0.02, 0.005])
Claim 2: plant peaks [$u_{\text{bl,ail}}$, $u_{\text{bl,rud}}$, p_s, β] vs [3, 2, 18, 0.5]	[2.55, 0.74, 15.33, **0.248**]	[2.52, 0.88, 15.37, **0.354**]	[2.61, 1.22, 15.44, **0.509**]	[2.65, 1.07, 15.39, **0.5002**]
Total command peaks $\lvert u\rvert$ [ail, rud] vs [3, 2]	[2.55, 1.19]	[2.52, 1.21]	[2.61, 1.22]	[2.65, 0.86]
(F): minimum margin over t (must be > 0)	[7.5, 0.37, 24.8, 0.034]	[7.9, 0.45, 24.9, 0.049]	—	—
Lemma 3: $\max_t(V - \bar{V}) \le 0$; $\bar{V}(40)/\bar{V}(0)$	$-4 \cdot 10^{-6}$; 0.93	$-1 \cdot 10^{-6}$; 0.95	$-4 \cdot 10^{-4}$; 0.97	$-4 \cdot 10^{-6}$; 0.94

Table 2. Simulation check of Theorem 2. "No tube" means the same reference-model CBF with $\delta \equiv 0$. `Robust_Adaptive_TK.m` (CBF on the plant, Theorem 1, $\gamma = 10^3$, Euler) gave peaks [2.61, 1.21, 15.44, 0.5005] for comparison.

All claims of Theorem 2 hold in both tube runs: the reference model stays in the shrunken box, the tube inequality holds, (F) stays positive, and the plant never leaves the original box. The tube is conservative. The actual $\lvert c_i e\rvert$ stays far below δ_i, and $\bar{V}$ shrinks by only 5–7% over 40 s because the gains do not converge. So β is held at 0.25° or 0.35° instead of 0.5°. Without the tube, the same reference-model design lets the plant cross the β limit (0.509° at $\gamma = 10^3$), which is exactly what your tube was meant to prevent. RK4 was used for these runs; I did not check whether your Euler step ($\Delta t = 10^{-4}$) is stable at $\gamma \ge 10^5$.

7. What this means for the code

- **If a guarantee is required**, use the Theorem 2 structure. This is your original structure with these changes:
  1. do not clamp δ;
  2. pick γ (with the observer-like RM, so a high γ does not oscillate) so that (F) holds;

3. add $\dot{\delta}_i$ and $C_{\lim} L_v e$ to a and b;
4. keep $x(0) = x_{\text{ref}}(0)$;
5. monitor (F) online.

Expect conservative behaviour, e.g. β held near $0.5^\circ - \delta_\beta$.

- **If small, bounded violations are acceptable**, `Robust_Adaptive_TK.m` (Theorem 1) tracks better and keeps violations of order $\sup |d_i|/\alpha_i$, which vanish as adaptation settles.
- **Scope of these proofs:** continuous time (Euler or RK4 adds a small discretization error); relative-degree-one rows; H_u invertible; the input bound is on u_{bl}, as in the papers.